

Document Chat RAG — AI Features & Roadmap

Recommended AI capabilities, enhancements, and implementation roadmap for the project.

1. Document Intelligence & Multi-Format Ingestion

- **Multi-Format Support:** Extend beyond PDF to ingest Word (.docx), PowerPoint (.pptx), Excel (.xlsx), CSV, Markdown, and plain text files using LlamaIndex connectors.
- **OCR & Scanned PDF Parsing:** Integrate vision/multimodal models (e.g., Llama 3.2 Vision, Tesseract, Marker) to reliably parse scanned PDFs, handwritten notes, and image-heavy pages.
- **Table & Chart Understanding:** Extract structured tabular data as Markdown/JSON representations for accurate numerical queries, financial tables, and calculations.
- **Multi-Document Indexing:** Support uploading and simultaneously indexing multiple documents with cross-document search, filtering, and comparative analysis.

2. Advanced Retrieval (Advanced RAG)

- **Hybrid Search (BM25 + Dense Vectors):** Combine keyword-based search with dense semantic embeddings to balance exact term lookup (IDs, product codes) with contextual meaning.
- **Reranking Layer:** Add cross-encoder rerankers (e.g., BAAI/bge-reranker-large or Cohere Rerank) to reprioritize retrieved chunks before sending to the LLM.
- **Query Transformation & HyDE:** Use Hypothetical Document Embeddings (HyDE) and query re-writing / sub-question decomposition to tackle ambiguous or multi-part queries.
- **Persistent Vector Database:** Connect to persistent vector databases (e.g., Qdrant, ChromaDB, Milvus) so uploaded documents persist across sessions without re-indexing.

3. Citations, Verification & Explainability

- **Exact Source Citations:** Display clickable references showing exact page numbers, source chunk text, and similarity confidence scores under each answer.
- **Interactive PDF Highlighting:** Clicking a citation jumps directly to and highlights the corresponding text excerpt within the embedded PDF preview pane.
- **Hallucination Guardrails & Fact-Checking:** Apply self-reflection checks or guardrail frameworks (e.g., NeMo Guardrails) to flag statements unsupported by the document context.

4. Automated Document Actions & Productivity

- **One-Click Executive Summary:** Generate automated executive summaries, TL;DR bullet points, and key takeaways immediately upon file upload.
- **Suggested Questions & Starter Prompts:** Analyze the document content to proactively generate 3-5 relevant starter questions for the user.
- **Quiz & Flashcard Generator:** Automatically generate study quizzes, multiple-choice tests, or flashcards from the uploaded document for training and onboarding.
- **Export Chat & Reports:** Enable exporting conversations, summaries, and Q&A; transcripts into downloadable PDF, Word, or Markdown reports with citations.

5. Multi-Agent & Conversational Capabilities

- **Agentic Tool Calling:** Equip the LLM with external tools (DuckDuckGo web search, Python code interpreter for calculations, database queries) when documents lack answers.
- **Dynamic Model Switcher:** Allow users to toggle between local Ollama models (Llama 3.2, Llama 3.3, Mistral, DeepSeek-R1) and external cloud APIs (OpenAI, Gemini, Anthropic).
- **Voice-to-Text & Text-to-Speech:** Incorporate Whisper speech recognition for voice input and TTS audio synthesis for spoken responses.

Implementation Roadmap

Phase	Key Deliverables	Impact & Complexity
Phase 1: Quick Wins	One-click document summary, auto-generated starter prompts, and source page citations.	High Impact / Low Effort
Phase 2: Core RAG Upgrade	Persistent vector store (Qdrant), multi-file uploads, and cross-encoder reranking.	High Impact / Medium Effort
Phase 3: Advanced Intelligence	Multimodal parsing (OCR/Vision), agentic web/code tool calling, and voice interaction.	High Impact / Advanced Effort